



SNo	Quiz + Test (Solutions with Scheme)	Marks
1.1	Green light has a wavelength 550nm. Calculate the potential difference through which an electron must be accelerated to have this wavelength. $V = \frac{(12.28)^2 \times 10^{-20}}{\lambda^2} = 5\mu V$	1
1.2	Calculate the zero-point energy that a neutron can have in a box of rigid walls of length $10^{-14}m$. $E = \frac{h^2}{8ma^2} = 3.285 \times 10^{-13} \text{ J or } 2.05 \times 10^6 \text{ eV.}$	1
1.3	Sketch (a) the wave function and (b) the probability density for the quantum state $n = 3$ for a particle in a one-dimensional potential well of infinite depth. $n = 3$.	2
1.4	Write the time independent Schrodinger wave equation and normalized wave function for a particle in 3-dimensional potential well. $\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} + \frac{8\pi^2 m E}{h^2} \Psi(x, y, z) = 0$ Normalized wave function is $\Psi = \sqrt{\frac{2}{a}} \sqrt{\frac{2}{b}} \sqrt{\frac{2}{c}} \sin \frac{n\pi}{a} x \sin \frac{n\pi}{b} y \sin \frac{n\pi}{c} z$	1 1
1.5	State Matthiesen's rule of resistivity in metals. It states that total resistivity of a metal is the sum of the resistivity due to phonon scattering, which is temperature dependent and the resistivity due to scattering by imperfections which is temperature independent.	1
1.6	Give one failure of classical free electron theory. Any one failure.	1
1.7	Define density of states in metals. It is defined as, the number of available states per unit volume per unit energy interval.	1
1.8	What is the probability of finding electron at energy level 7.2eV in Copper at 0K? Given the Fermi energy of Copper is 7.0 eV. Zero	1
1a	Explain de-Broglie's hypothesis of matter waves. Mention two reasons why de-Broglie waves are not electromagnetic waves? In nature energy manifests itself in two forms, namely matter and radiation.	



	• Nature loves symmetry. • As radiation can act like both wave and a particle, material particles (like electrons, protons, etc.) in motion should exhibit the property of waves. These waves due to moving matter are called matter waves or de-Broglie waves. 1. Matter waves are produced by charged or uncharged particles in motion. Whereas electromagnetic waves are produced only by a moving charged particle. Hence matter waves are non – electromagnetic waves. 2. In an isotropic medium the wavelength of an electromagnetic wave is a constant, whereas wavelength of a matter wave changes with the velocity of the particle. Hence matter waves are non- electromagnetic waves.	04
1b	A particle is moving in one dimensional potential box of infinite height and width $10\text{\AA}$. Calculate the probability of finding the particle within an interval of $4\text{\AA}$ at the centre of the box when it is in the state of least energy and first excited state. When $n=1$ $P = \int_{-2}^{2} \frac{2}{a} \sin^2\left(\frac{n\pi x}{a}\right) dx$ $P = \frac{1}{10} \left\{ 4 - \frac{5}{\pi} (\sin 252) - \sin 108 \right\}$ $P = \frac{1}{10} \left\{ 4 - \frac{5}{\pi} (-0.951 - 0.951) \right\}$ $P = \frac{1}{10} \{ 4 + 3.0286 \}$ $P = 0.7028 \text{ or } 70.28\%$ When $n=2$ $P=0.3064$ or 30.64%	02 02
2a	State and explain Heisenberg Uncertainty Principle. Hence, explain with the diagrams, the origin of spectral line broadening, and obtain an expression for the uncertainty in wavelength $\Delta\lambda$. Ans. According to the Heisenberg's uncertainty principle "It is impossible to specify precisely and simultaneously certain pairs of physical quantities like position and momentum that describe the behavior of an atomic system". Qualitatively, this principle states that in any simultaneous measurement the product of the magnitudes of the uncertainties of the pairs of physical quantities is equal to or greater than $h/4\pi$ (or of the order of h). Considering the pair of physical quantities such as position and momentum, we have $\Delta p \Delta x \geq h/4\pi$. Where Δp and Δx are the uncertainties in determining the momentum and the position of the particle.	02

When an atom absorbs a photon, it rises to the excited state and it will stay in the excited state for certain time called the lifetime. Lifetime of atoms in the excited levels is of the order of 10^{-8} s. When the atom comes to the ground state it emits a photon of energy exactly equal to the energy difference between the two levels as shown in the figure.

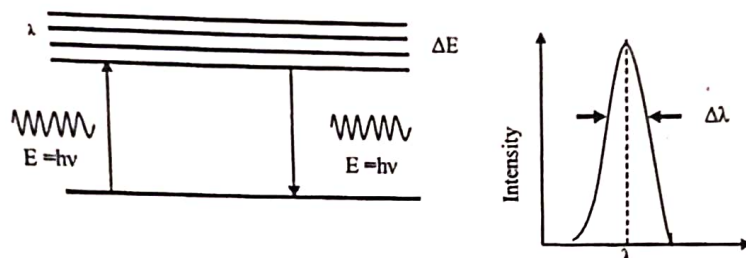


Fig. Line width for emitted photons

$$E = h\nu = \frac{hc}{\lambda} \dots \dots \dots (1)$$

The energy of the emitted photon is given by

Differentiating equation (1) with respect to wavelength (λ), we get

$$\Delta E = - \frac{hc \Delta \lambda}{\lambda^2}$$

$$|\Delta E| = \frac{hc \Delta \lambda}{\lambda^2} \dots \dots \dots (2)$$

According to Heisenberg's uncertainty principle, the finite lifetime Δt of the excited state means there will be an uncertainty in the energy of the emitted photon is given by

$$\Delta E \geq \frac{h}{4\pi \Delta t}$$

Substituting for ΔE from (2) and applying the condition of minimum uncertainty, we get

$$\frac{hc \Delta \lambda}{\lambda^2} = \frac{h}{4\pi \Delta t} \quad \Delta \lambda \geq \frac{\lambda^2}{4\pi c \Delta t}$$

2b An electron of energy 250 eV passes through an aperture such that the uncertainty in its position is 10^{-6} m. Using the Heisenberg uncertainty principle, determine the angle of emergence of the electron.

$$P = \sqrt{2mE} = 8.53 \times 10^{-24} \text{ kgms}^{-1}$$

$$\Delta p = 0.527 \times 10^{-28} \text{ kgms}^{-1}$$

$$\text{Angle of emergence} \Rightarrow \theta = \frac{\Delta p}{p} = 0.062 \times 10^{-4} \text{ rad} \quad 3.54 \times 10^{-4} \text{ degrees}$$

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3a	Explain normalization of wave function? Obtain the expression for normalized wave function for a particle in one dimensional potential well of infinite height. If ψ is a wave function associated with a particle, then $\psi^2 d\tau$ is the probability of finding the particle in a small volume $d\tau$. If it is certain that the particle is present in a volume τ then the total probability in the volume τ is unity i.e., $\int_{\tau} \psi^2 d\tau = 1$. This is the normalization condition. (with or without diagram) Normalization of wave function: Explanation of the infinite potential well with diagram and the wave function $\psi = B \sin kx = B \sin \frac{n\pi}{a} x$. The integral of the wave function over the entire space in the box must be equal to unity because there is only one particle within the box, the probability of finding the particle is 1. $\int_0^a \Psi ^2 dx = 1$ $B^2 \int_0^a \sin^2 \frac{n\pi}{a} x dx = 1$ $\frac{B^2}{2} \int_0^a \left(1 - \cos \frac{2n\pi x}{a}\right) dx = 1$ $\frac{B^2}{2} \left[\int_0^a dx - \int_0^a \cos \frac{2n\pi x}{a} dx \right] = 1$ $\frac{B^2}{2} \left[x - \frac{a}{2n\pi} \sin \left(\frac{2n\pi x}{a} \right) \right]_0^a = 1$ $\frac{B^2}{2} \left[a - \frac{a}{2n\pi} \sin 2n\pi - 0 \right] = 1$ $\therefore \frac{B^2 a}{2} = 1$ $\therefore B = \sqrt{\frac{2}{a}}$ Thus the normalized wave function of a particle in a one-dimensional box is given by, $\Psi_n = \sqrt{\frac{2}{a}} \sin \left(\frac{n\pi}{a} \right) x \quad \text{where, } n=1,2,3,\dots$	06 02 01 03
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3b

A particle is described by a wave function $\Psi = \sqrt{\frac{2}{L}} \sin \frac{n\pi}{L} x$ for $0 < x < L$ and $= 0$ elsewhere. Find the expectation value of the momentum of the particle.

$$\langle p \rangle = \int_0^L \Psi^*(x) \hat{p} \Psi(x) dx$$

Since $\Psi(x)$ is real $\Psi^*(x) = \Psi(x)$.

$$\text{Hence } \langle p \rangle = -i\hbar \int_0^L \Psi(x) \frac{d\Psi(x)}{dx} dx$$

$$\langle p \rangle = -i\hbar \int_0^L \left(\sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \right) \left(\sqrt{\frac{2}{L}} \frac{n\pi}{L} \cos \frac{n\pi x}{L} \right) dx$$

$$\langle p \rangle = -i\hbar \frac{2n\pi}{L} \int_0^L \sin \frac{n\pi x}{L} \cos \frac{n\pi x}{L} dx$$

$$\langle p \rangle = -i\hbar \frac{n\pi}{L} \int_0^L \sin(2 \frac{n\pi x}{L}) dx = 0$$

The particle has equal probability of momentum in +p and -p directions inside the box, so the average momentum is zero.

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4a

Describe the finite potential well with neat diagram and solve the time independent Schrödinger equation to obtain the wavefunctions for a particle in the well. Also explain the phenomenon of quantum tunnelling.

Description of the finite potential well with diagram.

Solving the time independent Schrödinger equation to obtain the wavefunctions for a particle in the well and matching the solutions at boundaries.

Explanation of Quantum tunnelling

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4b

In a solid, consider the energy level lying 0.01eV below Fermi level. What is the probability of this level not being occupied by an electron at 300K?

$$\text{Ans. } F(E) = 1 - \frac{1}{1 + e^{\frac{E - E_F}{kT}}} = \frac{1}{1 + e^{\frac{-\Delta E}{kT}}} = 0.405 \text{ or } 40.5\%$$

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5a

Mention any three assumptions of Quantum Free Electron theory. Prove that the probability of finding an electron of energy ΔE above the Fermi level is same as probability of not finding an electron at energy ΔE below the Fermi level.

Any three assumptions of QFET

$$F(E_F + \Delta E) = 1 - \{F(E_F - \Delta E)\}$$

$$\frac{1}{1 + e^{\frac{E_F + \Delta E}{kT}}} = 1 - \frac{1}{e^{\frac{E_F - \Delta E}{kT}}}$$

$$\frac{1}{1 + e^{\frac{\Delta E}{kT}}} = \frac{1}{1 + e^{\frac{\Delta E}{kT}}}$$

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5b

Using density of states function, derive an expression for electron concentration in metals at 0K. Number of free electrons /unit volume which possess energy in the range E and E+dE is given by $N(E) dE = g(E) \times dE \times f(E)$

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The number of free electrons/unit volume of the material, i.e., n is equal to the total number of electrons that are distributed in various energy levels upto E_F . Thus we have

$$n = \int_{E=0}^{E_F} N(E) dE$$

$$n = \int_{E=0}^{E_F} g(E) f(E) dE$$

But, $f(E) = 1$, at $T = 0K$

$$n = \int_{E=0}^{E_F} g(E) dE \times 1$$

$g(E) dE$ is given by, $g(E) dE = \frac{4\pi}{h^3} (2m)^{3/2} E^{1/2} dE$

$$n = \frac{4\pi}{h^3} (2m)^{3/2} \int_{E=0}^{E_F} E^{1/2} dE$$

$$n = \frac{4\pi}{h^3} (2m)^{3/2} \frac{2}{3} (E_F)^{3/2}$$

$$n = \frac{8\pi}{3h^3} (2m)^{3/2} (E_F)^{3/2}$$

This is the equation of concentration of electrons in a metal at $0K$.

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Date: 19.05.26	Semester: II	Quiz-II + Test - II	Max. Marks: 10+50
Academic Year 2025-2026 Course: Quantum Physics for Engineers (PY2211C)			

SNo	Questions PART A	Solution with scheme	Marks
1	What is the position of the Fermi level in an n-type semiconductor at $T = 0K$?	Halfway between the bottom of conduction band and donor level	1
2	At a normal metal-superconductor interface, what condition must the energy of an incident electron satisfy for Andreev reflection to occur?	Energy of the incident electron must be less than the superconducting energy gap of the superconductor	1
3	What percentage of normal electrons are present in a superconductor at absolute zero temperature and above the critical temperature?	At 0 K: Percentage of normal electrons is 0----- 1 mark Above critical temperature: Percentage of normal electrons is 100----- 1 mark	2
4	A Hall measurement is performed on a semiconductor bar with the magnetic field applied parallel to the direction of current. What is the Hall voltage measured across the sample? Justify the answer.	Hall voltage is zero----- 1 mark Since magnetic field and velocity of the charged carriers are parallel, Lorentz force on the carriers is zero----- 1 mark	2
5	A superconducting cylindrical wire of diameter 1 mm has a critical magnetic field of $4.3 \times 10^4 A/m$ at a temperature of 7.18 K. Calculate the critical current of the wire.	$I_c = 2\pi r H_c \approx 135 A$	1
6	What is Hall effect in materials?	The Hall Effect is the phenomenon in which a transverse electric field (perpendicular to both the current flow and an applied magnetic field) is developed across a conducting or semiconducting material when it carries an electric current in the presence of a perpendicular magnetic field.	1
7	"The Fermi level position of an intrinsic semiconductor remains unchanged with variation of temperature" – State whether this statement is true or false.	False	1
8	"The bandgap of a semiconductor narrows with very high doping" – State whether this statement is true or false.	True	1

SNo	Questions PART B	Solution with scheme	Marks
1a	Derive an expression for the electron concentration in the conduction band of an intrinsic semiconductor at a temperature T.	$n = \int_{E_c}^{\infty} f(E)g(E)dE$ and in the conduction band $g(E) = \frac{4\pi}{h^3} (2m_e^*)^{3/2} (E - E_c)^{1/2} \dots\dots$ with explanation----- 2 marks Substitution in the integral with the approximation, $\frac{1}{(1+\exp(\frac{E-E_F}{KT}))} \approx \exp(-\frac{E-E_F}{KT})$ ----- 1 mark	7



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		From change of variables to $n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \exp \frac{E_F - E_C}{KT} \int_0^\infty x^{1/2} e^{-ax} dx$ ----- 2 marks Up to final answer, $n = 2 \left(\frac{2\pi m_e^* KT}{h^2} \right)^{3/2} \exp \left(\frac{E_F - E_C}{KT} \right)$ ---- 2 marks	
1b	For an intrinsic semiconductor with band gap $E_g = 0.7$ eV, calculate the concentration of intrinsic charge carriers at 300 K. Assume the effective mass of the electron to be the same as the rest mass of the electron.	$E_g = 0.7$ eV = 1.12×10^{-19} J $T = 300$ K, $m^* = m_0 = 9.109 \times 10^{-31}$ kg In intrinsic semiconductor, $n = p$. Let us calculate n. ----- 1 mark $n = 2 \left(\frac{2\pi m_e^* KT}{h^2} \right)^{3/2} \exp \left[\frac{-(E_C - E_F)}{KT} \right] =$ $2 \left(\frac{2\pi m_e^* KT}{h^2} \right)^{3/2} \exp \left[\frac{-E_g}{2KT} \right] = 3.38 \times 10^{19} m^{-3} = p$ ----- - 2 marks	3
2a	With the help of a labelled graph and band diagram, explain the variation of electron concentration with temperature in an n-type semiconductor. Explain all the regions.	Labelled graph ----- 1 mark Explanation of ionization region + Band diagram ----- 2 marks Explanation of depletion region + Band diagram ----- 2 marks Explanation of intrinsic region + Band diagram ----- 2 marks	7
2b	A semiconductor has a band gap of 1.2 eV. The effective mass of holes is 6 times the effective mass of electrons. The Fermi energy is found to be 28 meV above the midgap energy level. Find the temperature of the semiconductor.	$E_F = \left(\frac{E_C + E_V}{2} \right) + \frac{3}{4} KT \ln \left(\frac{m_h^*}{m_e^*} \right)$ $= E_{mid} + \frac{3}{4} KT \ln \left(\frac{m_h^*}{m_e^*} \right)$ Shift from middle of the bandgap = 28×10^{-3} eV $1.6 \times 10^{-19} = \frac{3}{4} KT \ln \left(\frac{m_h^*}{m_e^*} \right)$ ----- 1 mark $m_h^* = 6m_e^*$ Solving, $T \approx 242$ K ----- 2 marks	3
3a	Explain BCS theory of superconductors highlighting the Cooper pair formation and phase coherent state.	Definition of Cooper pairs which do not get scattered --- ----- 1 mark How the Cooper pairs are formed due to the lattice effect despite having similar charges. How temperature affects electron-phonon coupling ----- 3 marks The form of the Cooper pair wavefunction with constant phase throughout the sample. Essential to have zero resistant state ----- 2 marks	6
3b	A superconducting niobium wire and a copper wire of resistance R are connected in parallel. The circuit is powered by a dc battery of voltage 5 V. What is the power dissipated in the circuit? Explain your answer. What will the voltage drops be across the niobium and copper wire if they are connected in series?	Superconducting Nb has no resistance. It will act as short and entire current will flow through this branch without any dissipation ----- 2 marks If they are in series, still resistance of Nb is zero. So, voltage drop across Nb wire is zero. 1 mark The entire applied voltage will be dropped across the copper wire. $V_{Cu} = 5$ V ----- 1 mark	4



	In both the above cases, the circuit is maintained below the critical temperature of niobium.		
4a	What is Meissner effect? How do Cooper pairs contribute to this effect? Show that the magnetic susceptibility of type-I superconductors is -1.	Definition of Meissner effect - complete expulsion of magnetic field + Diagram ----- 2 marks Contribution - Circulating Cooper pairs creating its own field and cancelling the external field ----- 2 marks From $\vec{B} = \mu_0(\vec{H} + \vec{M})$ with explanation of terms till $\chi = -1$ ----- 2 marks	6
4b	A type-I superconductor has a critical temperature of 3.72 K and its maximum critical field is 2.46×10^4 A/m. Find the critical field of the material at 2 K. If an external field is applied at 2 K creating a magnetic intensity $\vec{H} = 1.43 \times 10^4$ A/m inside the superconductor, find the magnetic induction $\vec{B}$ and magnetization $\vec{M}$ deep inside the superconductor.	Critical temperature: $T_c = 3.72$ K Maximum critical field: $H_c(0) = 2.46 \times 10^4$ A/m Temperature of interest: $T = 2$ K $H_c(T) = H_c(0) \left[1 - \left(\frac{T}{T_c} \right)^2 \right] = 1.75 \times 10^4$ A/m ----- 2 marks As 1.43×10^4 A/m is below the critical field at 2 K. So, $\vec{B} = \vec{0}$ inside ----- 1 mark $\vec{B} = \mu_0(\vec{H} + \vec{M}) = \vec{0}$ $\Rightarrow \vec{M} = -\vec{H} = -1.43 \times 10^4$ A/m ----- 1 mark	4
5a	What is ac Josephson Effect? Explain with a diagram. Obtain the expression of the angular frequency of the emitted electromagnetic radiation in an ac Josephson effect. Write one application of DC SQUIDS.	Mention of Josephson junction - thin insulating barrier between two superconducting pads - Application of dc Voltage causing time evolution of phase difference between the wavefunctions of two superconducting sides - Equation $\frac{d\delta}{dt} = \frac{2eV}{\hbar}$ ----- 2 marks Diagram ----- 1 mark $I = i_c \sin \delta = i_c \sin \left(\frac{2eV}{\hbar} t \right) = i_c \sin(\omega t)$ Where $\omega = \frac{2eV}{\hbar}$ is the angular frequency of emitted electromagnetic radiation ----- 2 marks Application: As very sensitive magnetometer (or any one of the listed applications) ----- 1 mark	6
5b	Consider a p-type semiconductor in form of a rectangular bar. The longest side is kept along x-axis. The width (parallel to y-axis) and thickness of the sample are 3mm and 1 mm respectively. The carrier concentration in the bar is $5 \times 10^{21} \text{ m}^{-3}$. A current $I = 10$ mA flows in the positive x-direction and a field $B = 0.5$ T is applied in the positive z-direction. Calculate the Hall coefficient and the Hall voltage.	Semiconductor type: p-type Width (y-direction): $w = 3 \text{ mm} = 3 \times 10^{-3} \text{ m}$ Thickness (z-direction): $t = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$ Carrier concentration: $p = 5 \times 10^{21} \text{ m}^{-3}$ Current: $I = 10 \text{ mA} = 10 \times 10^{-3} \text{ A}$ Magnetic field: $B = 1 \text{ T}$ (in +z direction) Charge of hole: $e = 1.6 \times 10^{-19} \text{ C}$ 6.25 mV	4

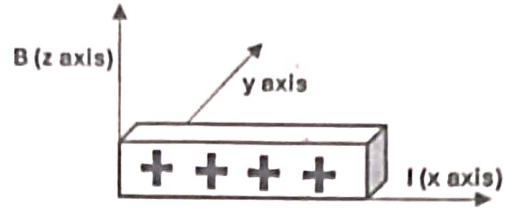


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Indicate which face of the sample is at higher potential in a neat diagram.

$$R_H = \frac{1}{pe} = 1.25 \times 10^{-3} \text{ m}^3/\text{C}$$
$$V_H = 6.25 \text{ mV} \text{ ----- 2 marks}$$



Holes will be deflected along -y direction. The face which is at lower y value will be at higher potential-----
- 1 mark

Labelled diagram ----- 1 mark